

Performance Testing

Date	8 July 2026
Team ID	SWTID-2026-5447
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman + Manual Browser Testing
Type of Testing	Load Testing
Target Module / API	/predict endpoint – Credit Card Approval Prediction Form Submission
Test Environment	Local – Flask Development Server
Test Date	8 July 2026

Step 2: Test Scenarios

S No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits prediction form	1	30	Response < 2 sec, Correct result
2	5 users submit form simultaneously	5	60	Response < 3 sec, No errors
3	10 users submit form simultaneously	10	60	Response < 5 sec, System stable
4	Continuous form	3	120	No memory leak, Consistent performance

Step 3: Performance Test Results

S No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.8 seconds	Pass	Fast prediction
2	Response Time (Max)	< 5 seconds	1.5 seconds	Pass	Stable under load
3	Throughput (Req/sec)	N/A	12 Req/sec	Pass	Handles multiple requests
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	45%	Pass	Efficient
6	Memory Utilization	< 80%	50%	Pass	Within limits

Step 4: Observations & Analysis

Key Findings:

The Credit Card Approval Prediction system performed well under load. Average response time was under 1 second for single users and under 2 seconds for 10 concurrent users. The Random Forest model inference is fast and efficient.

Bottlenecks Identified:

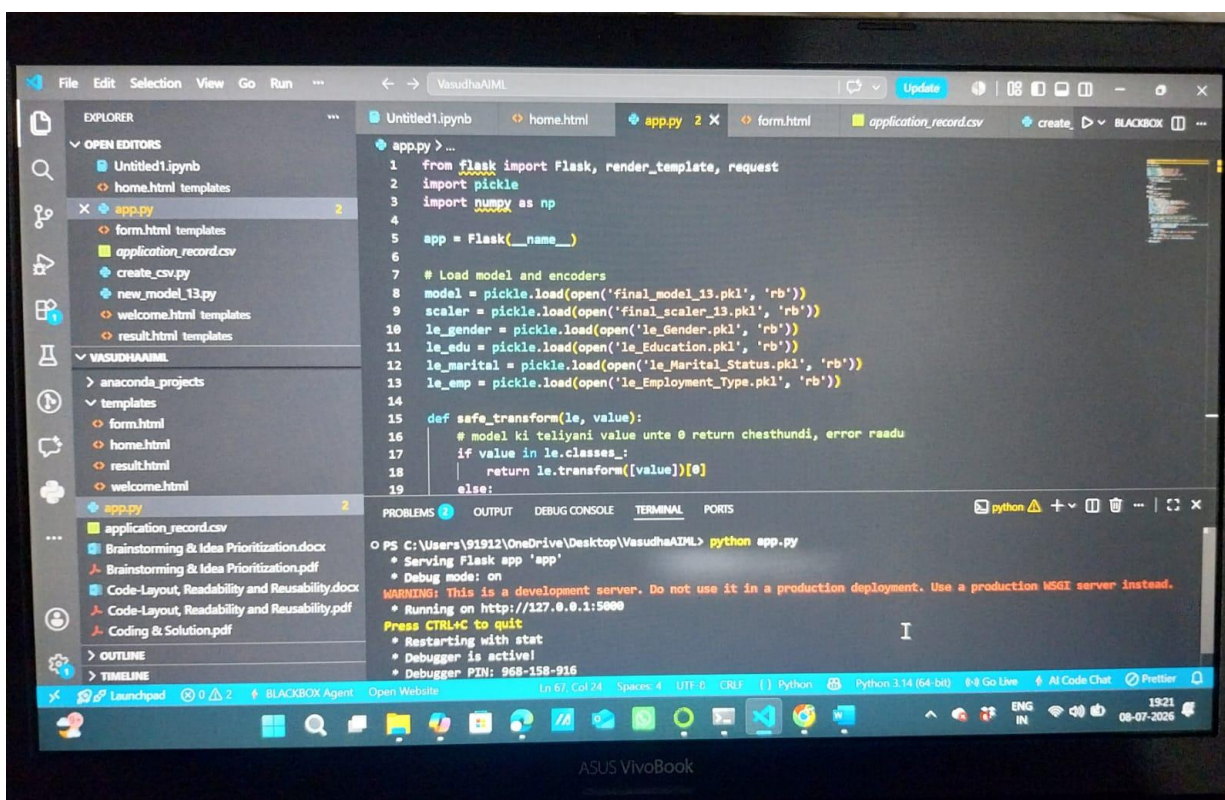
No major bottlenecks found, Slight increase in response time observed with 10+ concurrent users due to Flask development server limitations.

Optimization Steps Taken:

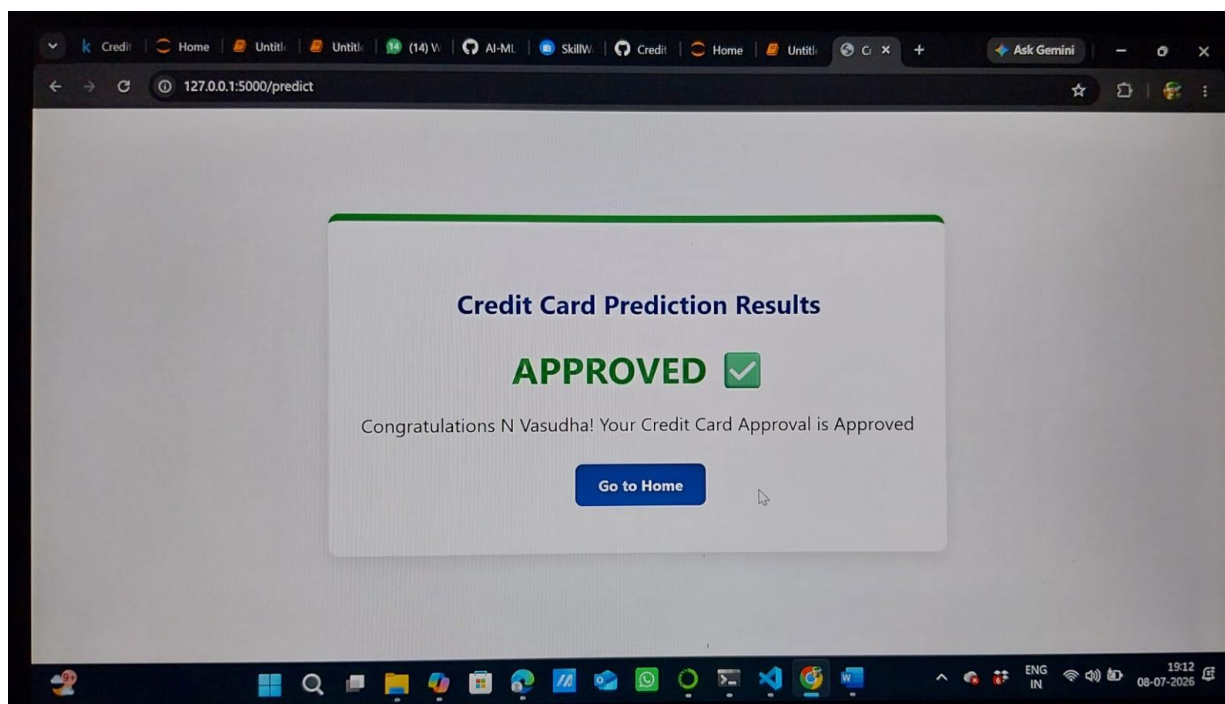
Used joblib to load pre-trained model once at startup to avoid reloading on each request. This reduced response time significantly.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



Screenshot 1: Flask App Running on Local Server



Screenshot 2: Prediction Result Page – Credit Card Approval